RUDRAEDGE AEROSPACE

Project - Rudra Edge RE-T1(Spectra)
Daily Engineering Log

1. TEAM MEMBER INFORMATION

Engineer Name: Suraj Nadda
Designation / Role: Software & Data Engineer
Task Order Reference: 14 October 2026 – Performance benchmarking

2. DAILY LOG ENTRY FORM

- **Date:** 14 October 2026

- **1. Assigned Task:**

  Performance benchmarking: loop vs vectorized throughput analysis

- **2. What Was Studied:**

  Studied Python loop-based processing and NumPy vectorized operations for numerical data processing. Reviewed execution-time measurement, average runtime, processing throughput and speedup factor. Also studied how to compare numerical outputs to verify correctness and how benchmarking helps evaluate processing efficiency.

- **3. What Was Implemented:**

  Implemented a Python benchmark to process 200,000 numerical elements using a Python loop and NumPy vectorization. Measured execution time across three runs and calculated average runtime, throughput and speedup. Compared both outputs using NumPy and saved the results in a structured JSON report.

- **4. Tools / Libraries Used:**

  VS Code, PowerShell, Python 3.14.6, JSON, Pathlib ,

- **5. Output File Name-** suraj_rudra_task/ Day10/codes/performance_benchmark.py
  Path-https://github.com/surajnadda29/suraj_rudra_t1.git

- **6. Problems Encountered:**
  No problem was encountered

- **7. How Problem Was Resolved:**
  No problem was encountered.

- **8. Next Task Scheduled:**

  (15 October 2026)- Telemetry Fault recovery – corrupted/missing telemetry handling and interpolation logic

- **9. Personal Expenditure: ₹0 (Verified Mandate)**

3. ENGINEER SELF-DECLARATION & SIGN-OFF

I hereby formally declare that today's technical progression, synthetic dataset code, and local repository paths have been documented truthfully in compliance with Section 6 (Common Technical Standards). I certify with absolute transparency that my personal financial expenditure for today's assigned task is exactly ₹0, utilizing strictly local open-source infrastructure.

Candidate/Engineer Signature:  Date: 14/10/2026

Full Name: Suraj Nadda

PROJECT LEAD VERIFICATION BOX — TO BE FILLED BY SARVESH SHARMA ONLY

I have reviewed today's technical output, local repository commits, and verified the project compliance standards for this day's assigned task.

Task Evaluation Status:

[ ] Approved / Completed

[ ] Needs Revision / Delayed

[ ] Technical Block Encountered

Operational Expenditure Confirmed: ₹0 (Strict Zero-Cost Compliant)

Remarks/Notes: _____

Project Lead Signature: _____ Date: ____/____/2026